

Duality: Lecture Notes

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Definition 0.1 (Linear Functional). A linear functional on V is a linear map from V to F ; that is, an element of $\mathcal{L}(V, F)$.

Definition 0.2 (Dual Space). The dual space of V , denoted by V' , is the vector space of all linear functionals on V . In other words $V' = \mathcal{L}(V, F)$.

Theorem 0.1. *Suppose V is finite-dimensional. Then V' is also finite-dimensional and*

$$\dim V' = \dim V.$$

Proof. $\dim \mathcal{L}(V, F) = (\dim V)(\dim F) = \dim V$. □

Definition 0.3. Let $v_1, \dots, v_n$ be a basis of V . Then the dual basis of V is the list $\varphi_1, \dots, \varphi_n$ of linear functionals on V s.t.

$$\varphi_j$$